



POCKETSMART AI • PHASE 3

Project Design Phase

Designing the architecture, modules, data flow, and user interface

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1. High-Level Architecture

The guide describes a modular platform with a frontend, a FastAPI/Flask-style backend layer, Gemini 1.5 Flash Pro for AI processing, and third-party sources for product/service data. The later implementation sections specifically document FastAPI and Jinja2 templates.

2. Main Components

Component	Purpose
Frontend UI	HTML/CSS/Jinja2 interface for budgets, preferences, images, planner forms, and recommendation display.
Backend	FastAPI implementation that handles routes, authentication, sessions, structured inputs/outputs, CORS, and communication with the AI layer.
Gemini AI Layer	Processes budget context, user preferences, text inputs, and optional images to produce domain-specific recommendations.
External Platforms	Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO and similar sources are referenced for product/service data.

3. Backend Route Design

- /generate-home
- /generate-party
- /generate-jewelry
- /register
- /login
- /logout
- /token
- /session-info
- /session-data
- /recommendations-details
- /history
- /startup

4. Frontend Design

The documented UI uses HTML/CSS and Jinja2 templates, with forms connected to FastAPI POST routes and recommendation results displayed in structured, card-like layouts.

Phase Outcome

The design separates presentation, API routing, AI/recommendation logic, authentication/session handling, and external data sources so the application remains modular.

Source basis: supplied PocketSmart AI project guide. The reference YouTube video was provided by the user but could not be fetched in this environment, so no video-specific claims have been added.